

Math Camp

Lecture 9

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Optimization

- A central tenet — perhaps *the* central tenet — of economics is that agents make decisions in a manner consistent with optimizing an objective
- Most interesting optimization problems in economics involve constraints
 - Scarcity is a central theme after all
- The principal analytic tool for elementary optimization problems is the Kuhn–Tucker theorem
- Besides stating the theorem, we use the total differential to develop the intuition underlying it
 - This *variational approach* extends to many other types of optimization problems, including continuous-time optimal control

Background notions

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 and $K \subset \mathbb{R}^n$
 - f is the *objective function* and K the *constraint set*
 - When $K = \mathbb{R}^n$ the problem is *unconstrained*
- $\mathbf{x}^* \in K$ is a *local maximum* provided there is some $\varepsilon > 0$ such that for all $\mathbf{x} \in K \cap B(\mathbf{x}^*, \varepsilon)$ it follows that $f(\mathbf{x}^*) \geq f(\mathbf{x})$
- It is a *global maximum* provided $\mathbf{x} \in K \implies f(\mathbf{x}^*) \geq f(\mathbf{x})$
- To find the global maximum the standard approach is to first identify all local maxima and then pick the biggest one

Background notions

- It is common for both f and K to depend on a vector of parameters $\eta \in \mathbb{R}^m$
- The standard presentation of the problem takes two forms

$$V(\eta) = \max_{\mathbf{x} \in K(\eta)} f(\mathbf{x}, \eta)$$

$$\mathbf{x}^*(\eta) = \arg \max_{\mathbf{x} \in K(\eta)} f(\mathbf{x}, \eta)$$

- $V(\eta)$ is the value attained by f at the maximum
- $\mathbf{x}^*(\eta)$ is the vector in K that realizes the maximum

Existence

- Everything that follows characterizes a maximum *assuming there is one*
 - The first-order conditions are necessary: they say what must be true at $\mathbf{x}^*$, not that $\mathbf{x}^*$ is there to be found
 - So the first step in any problem is to establish that a solution exists
- This is settled by Weierstrass: if f is continuous and K is compact, then f attains both a maximum and a minimum on K
 - Continuity and compactness are cheap to check, and between them they rule out everything that can go wrong

Existence

- If $K = (0, 1)$ and $f(x) = x$, the constraint set is not closed and there is no maximum
- If $K = [0, \infty)$ and $f(x) = x$, it is not bounded and again there is no maximum
- When K depends on a parameter the corresponding statement is Berge's *theorem of the maximum*: if f is continuous and $K(\eta)$ is compact and varies continuously, then V is continuous
 - It is the result that licenses treating V as a well-behaved object

Interior and boundary

- The *interior* of $A \subset \mathbb{R}^n$, written A° , is the largest open set contained in A
- The *boundary* of A , written ∂A , comprises those points in A that are not in A° , i.e. $\partial A = A \setminus A^\circ$
 - If $A = [a, b] \subset \mathbb{R}$ then $A^\circ = (a, b)$ and $\partial A = \{a, b\}$
 - The interior may be empty: if $A = \{a\} \subset \mathbb{R}$ then there are no points in A°
 - A is open if and only if $A = A^\circ$

Interior and corner solutions

- K will often be taken as compact, i.e. closed and bounded, and will very often have a non-empty interior
- In this case there are two types of solutions
 - *Interior solutions*, found in K°
 - *Corner solutions*, found on the boundary, in ∂K
- If $K = [a, b]$ then an interior solution is $x^* \in (a, b)$ and a corner solution is $x^* \in \{a, b\}$
- The conditions characterizing interior and corner solutions are typically distinct, and both types must be checked

Necessary and sufficient conditions

- Two types of conditions are used to characterize a solution, be it interior or corner
 - *Necessary* conditions are usually called first-order conditions, because they involve first-order local approximations of the objective
 - The FOC provide statements that are true about the solution $\mathbf{x}^*$ assuming it exists
 - *Sufficient* conditions are usually called second-order conditions, for analogous reasons
 - The SOC provide conditions that guarantee existence

The univariate case

- The univariate case provides a great deal of intuition for the much richer multivariate environment
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $K = [a, b]$
- To derive the first-order conditions we always start the same way
 - ▶ Assume x^* is a local maximum, and use total differentials to see what must be true about it
 - ▶ Start at a local maximum and consider all feasible differential variations
 - ▶ These variations cannot increase the value of the objective — if they did, you would not have been at a local maximum

The univariate case

- More formally, if x^* is a local maximum then $df = f'(x^*) dx \leq 0$
 - If $df = f'(x^*) dx > 0$ then $f(x^* + dx) > f(x^*)$, contradicting that x^* is a local maximum
- This “no-arbitrage argument” is at the heart of variational arguments
- First suppose x^* is an *interior* local maximum
 - We could have considered both positive variations, $dx > 0$, and negative variations, $dx < 0$
 - If $df = f'(x^*) dx \leq 0$ for both then it must be that $f'(x^*) = 0$
- Now turn to a *corner*, and suppose $x^* = a$
 - The only feasible differential variation is $dx > 0$, so $f'(x^*) \leq 0$
 - Symmetrically, if $x^* = b$ then $f'(x^*) \geq 0$

The univariate case

The univariate case

- Now second-order conditions. Suppose we have found all points in $[a, b]$ satisfying the FOC.
 - Which of these really are local maxima?
- Start at the corners. Suppose $f'(a) \leq 0$
 - Since $dx > 0$ implies $df = f'(a) dx \leq 0$, for $\varepsilon > 0$ small enough, if $x \in (a, a + \varepsilon)$ then $f(x) \leq f(x^*)$
 - No additional conditions are needed at either corner
- Now a potential interior solution. That $f'(x^*) = 0$ is not enough
 - It could be a minimum, or a horizontal point of inflection, e.g. x^3 at the origin

The univariate case

- Consider the second-order Taylor expansion around the candidate optimum

$$\begin{aligned} f(x) &\approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 \\ &= f(x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 \end{aligned}$$

where the equality follows from $f'(x^*) = 0$

- Near x^* , provided $f''(x^*) \neq 0$, the function f looks like a parabola
 - x^* is a local maximum if this parabola faces downward, i.e. if the coefficient on x^2 is negative
 - Since this coefficient is $f''(x^*)$, the candidate really is a local maximum provided $f''(x^*) < 0$

The univariate case

$$f = x^2 - 3x + 2 \text{ on } \left[\frac{4}{3}, 3\right]$$

The multivariate case

- The multivariate case is more difficult principally because the constraint sets can be much more complicated
- We begin with the unconstrained problem, since links nicely to the univariate case
- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $K = \mathbb{R}^n$, and assume $\mathbf{x}^*$ is a local maximum
 - Contemplate a differential variation $d\mathbf{x}$: it must be that $df = Df(\mathbf{x}^*) d\mathbf{x} \leq 0$
 - $Df(\mathbf{x}^*)$ is the row of partials $f_i(\mathbf{x}^*)$
- Since $K = \mathbb{R}^n$, any variation is feasible. Fix i and vary only along the i th coordinate axis
 - Then $df = f_i(\mathbf{x}^*) dx_i$, and since dx_i could be positive or negative, $f_i(\mathbf{x}^*) = 0$
 - This must be true for all i

Counting conditions

- An important observation is warranted
 - There are exactly n first-order conditions, $f_i(\mathbf{x}^*) = 0$ for $i = 1, \dots, n$
 - And there are, in effect, n unknowns, the x_i^*
- The number of FOC corresponds to the degrees of freedom you have in selecting potential variations
 - In the unconstrained case, that is the number of choice variables
- This insight carries forward to the constrained case, and can be useful for determining when you have a complete set of FOC

Definiteness

- The second-order term of the Taylor expansion is a quadratic form in the variation $\mathbf{v} = \mathbf{x} - \mathbf{x}^*$, so its sign is what decides the matter
- Recall that for an $n \times n$ symmetric matrix A
 - ▶ *Positive definite*: $\mathbf{v}' A \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$; all eigenvalues positive
 - ▶ *Negative definite*: $\mathbf{v}' A \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$; all eigenvalues negative
 - ▶ *Indefinite*: $\mathbf{v}' A \mathbf{v}$ takes both signs; eigenvalues differ in sign
 - ▶ The *semidefinite* cases weaken the strict inequalities, and permit zero eigenvalues
- The Hessian $D^2 f(\mathbf{x}^*)$ is symmetric by Young's theorem, so all of this applies to it

Definiteness

$$N = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

The multivariate case

- As in the univariate case the sufficient conditions derive from the second-order Taylor expansion

$$\begin{aligned} f(\mathbf{x}) &\approx f(\mathbf{x}^*) + Df(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)' D^2 f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) \\ &= f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)' D^2 f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) \end{aligned}$$

- Suppose $D^2 f(\mathbf{x}^*)$ is negative definite. Then, provided $\mathbf{x}$ is near $\mathbf{x}^*$,

$$f(\mathbf{x}^*) - f(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \mathbf{x}^*)' D^2 f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) > 0$$

- That is, $\mathbf{x}^*$ is a local maximum

Concavity and global maxima

- Everything so far is *local*: to find the global maximum we must still collect all the local ones and pick the biggest
- Concavity removes that step
- f is *concave* on a convex set U provided that for all $\mathbf{x}, \mathbf{y} \in U$ and $\alpha \in (0, 1)$

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \geq \alpha f(\mathbf{x}) + (1 - \alpha) f(\mathbf{y})$$

- f is *convex* if the inequality is reversed

Concavity and global maxima

- Let $U \subset \mathbb{R}^n$ be open and convex and $f : U \rightarrow \mathbb{R}$ be C^2
 - f is concave on U if and only if $D^2 f(\mathbf{x})$ is negative semidefinite for every $\mathbf{x} \in U$
 - f is convex on U if and only if $D^2 f(\mathbf{x})$ is positive semidefinite for every $\mathbf{x} \in U$
- If f is concave on U and $Df(\mathbf{x}^*) = 0$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a *global* maximum of f on U
 - Symmetrically, if f is convex and $Df(\mathbf{x}^*) = 0$ then $\mathbf{x}^*$ is a global minimum
- This is why so many economic models assume a concave objective: the FOC alone then deliver the solution

Equality constraints

- Constraints in the multivariate case commonly come in two forms: equality and inequality constraints
- Equality constraints present as the solution set to a system of equations

► Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^m$. Then

$$K = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = \mathbf{c}\}$$

- We could always rewrite g to set $\mathbf{c} = \mathbf{0}$, but it is often convenient in applications to have non-zero $\mathbf{c}$
- g may be interpreted as a column vector of m functions $g^i : \mathbb{R}^n \rightarrow \mathbb{R}$

Equality constraints

- Same approach: assume $\mathbf{x}^*$ is a local maximum, so for any feasible variation $df = Df(\mathbf{x}^*) d\mathbf{x} \leq 0$
- But this time not all variations are feasible
 - We may only choose differentials so that $\mathbf{x}^* + d\mathbf{x}$ remains in K
 - How do we do this? The implicit function theorem
- Suppose $m = 1$ and $n = 2$. To ensure the constraint continues to bind, $Dg(\mathbf{x}^*) d\mathbf{x} = 0$, i.e.

$$g_1(\mathbf{x}^*) dx_1 + g_2(\mathbf{x}^*) dx_2 = 0$$

or

$$dx_2 = -(g_2(\mathbf{x}^*))^{-1} g_1(\mathbf{x}^*) dx_1$$

provided $g_2(\mathbf{x}^*) \neq 0$

Equality constraints

- We conclude that

$$\begin{aligned}df &= f_1(\mathbf{x}^*) dx_1 + f_2(\mathbf{x}^*) dx_2 \\&= \left[f_1(\mathbf{x}^*) - (g_2(\mathbf{x}^*))^{-1} g_1(\mathbf{x}^*) f_2(\mathbf{x}^*) \right] dx_1 \leq 0\end{aligned}$$

- Since there is no sign restriction on dx_1 ,

$$\frac{f_1(\mathbf{x}^*)}{g_1(\mathbf{x}^*)} = \frac{f_2(\mathbf{x}^*)}{g_2(\mathbf{x}^*)}$$

- So with $m = 1$ and $n = 2$ there is one FOC and one constraint
 - A single FOC even though there are two choice variables
 - This reflects that, because of the constraint, there is only one degree of freedom available when choosing the variation

Tangency

- The condition may be rewritten

$$\frac{f_1(\mathbf{x}^*)}{f_2(\mathbf{x}^*)} = \frac{g_1(\mathbf{x}^*)}{g_2(\mathbf{x}^*)}$$

- Setting the total differential of f to zero, the slope of the level curve of f through $\mathbf{x}^*$ is $-f_1(\mathbf{x}^*)/f_2(\mathbf{x}^*)$
- Setting the total differential of g to zero, the slope of the constraint set at $\mathbf{x}^*$ is $-g_1(\mathbf{x}^*)/g_2(\mathbf{x}^*)$

Tangency

- So at a constrained maximum the highest attainable level curve of f is *tangent* to the constraint set
 - If it were not, the level curve would cut through the constraint set, and moving along the constraint in one direction or the other would raise f
- Equivalently, $Df(\mathbf{x}^*) = \lambda Dg(\mathbf{x}^*)$: the two derivatives line up at $\mathbf{x}^*$
 - Each is perpendicular to the curve it is associated with
 - And at $\mathbf{x}^*$ those two curves share a tangent line

The method of Lagrange

- The same analysis can be completed for general m and n , but it becomes quite tedious
- Thankfully there is an easier way to reach the same FOC
 - Add variables — multipliers — which convert the constrained problem into an unconstrained one
 - Then use the FOC already developed
- Write

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda'(\mathbf{c} - g(\mathbf{x}))$$

- $\mathcal{L}$ is the *Lagrangian*
- $\lambda \in \mathbb{R}^m$ is a vector of unknowns — really they are dual, or co-variables — called the *Lagrange multipliers*
- There is one multiplier for each constraint

The method of Lagrange

Method of Lagrange

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are C^2 , if $\mathbf{x}^* \in \mathbb{R}^n$ is a local maximum of f subject to the constraint $g(\mathbf{x}^*) = \mathbf{c}$ and if $\text{rank}(Dg(\mathbf{x}^*)) = m$, then there exists $\lambda^* \in \mathbb{R}^m$ such that

- $\mathcal{L}_{x_i}(\mathbf{x}^*, \lambda^*) = 0$ for $i = 1, \dots, n$
- $\mathcal{L}_{\lambda_j}(\mathbf{x}^*, \lambda^*) = 0$ for $j = 1, \dots, m$

The method of Lagrange

- This provides a mechanical procedure for solving a maximization problem with equality constraints for which you know a solution exists
 - Check the constraint qualification $\text{rank}(Dg(\mathbf{x}^*)) = m$
 - Write down the Lagrangian
 - Take all the first partials and set them equal to zero
 - Solve for the potential $\mathbf{x}^*$ and λ^*
- Among them is the solution you are looking for

The method of Lagrange

- Returning to $n = 2$ and $m = 1$

$$\mathcal{L}_{x_1} = 0 \implies f_{x_1} - \lambda g_{x_1} = 0$$

$$\mathcal{L}_{x_2} = 0 \implies f_{x_2} - \lambda g_{x_2} = 0$$

$$\mathcal{L}_\lambda = 0 \implies g(\mathbf{x}) = c$$

- Combining the first two equations

$$\lambda = \frac{f_1(\mathbf{x}^*)}{g_1(\mathbf{x}^*)} = \frac{f_2(\mathbf{x}^*)}{g_2(\mathbf{x}^*)}$$

reproducing the earlier condition

- The last equation simply reproduces the constraint

Counting conditions again

- Recall the count in the unconstrained problem: n first-order conditions for n choice variables
- With m equality constraints the Lagrangian has $n + m$ arguments, and Lagrange's method delivers $n + m$ equations
 - n of them from $\mathcal{L}_{x_i} = 0$
 - m of them from $\mathcal{L}_{\lambda_j} = 0$, which simply reproduce the constraints
- And there are $n + m$ unknowns, the x_i^* and the λ_j^*
- The count works out because each constraint costs a degree of freedom in choosing the variation, but buys back a multiplier